Fundamental Modules - 2

WEEK 2: ENGINE SUPPORT & POWER DELIVERY

Days 7-12 | Detailed Hour-by-Hour Structure

Week Theme: *"Keeping the engine alive and transmitting power to the wheels"*

DAY 7: EMISSIONS CONTROL SYSTEMS + BASIC FLUID CHECKING

Learning Objectives:

- Understand emission standards and environmental responsibility
- Identify emission control components on vehicles
- Master basic fluid level checking procedures
- Understand fluid types and specifications

MORNING SESSION: THEORY (3 Hours)

Hour 1 (9:00 AM - 10:00 AM): Emission Standards & Environmental Impact

Why Emissions Matter:

- Air pollution and health effects
- Climate change contribution
- Legal requirements (government standards)
- Social responsibility of technicians

Harmful Emissions:

- **Carbon Monoxide (CO):**
  - Incomplete combustion
  - Toxic to humans
  - Rich fuel mixture creates more CO
- **Hydrocarbons (HC):**
  - Unburned fuel
  - Smog formation

    - Evaporative emissions from fuel system
- **Nitrogen Oxides (NOx):**
    - Formed at high temperatures
    - Respiratory irritant
    - Acid rain contributor
    - Smog formation
- **Particulate Matter (PM):**
    - Soot from diesel engines
    - Respiratory damage
    - Visible black smoke
- **Carbon Dioxide (CO2):**
    - Greenhouse gas
    - Climate change
    - Not toxic but environmental concern

Indian Emission Standards:

- **BS4 (2017-2020):**
    - Basic catalytic converter
    - OBD-II required
    - Moderate emission limits
- **BS6 (2020-present):**
    - 5x stricter than BS4
    - Real Driving Emissions (RDE) testing
    - Particulate filters mandatory
    - Better fuel quality required

Global Standards (Awareness):

- Euro 6 (Europe)
- EPA Tier 3 (USA)
- China 6
- BS6 based on Euro 6

Hour 2 (10:00 AM - 11:00 AM): Emission Control Technologies

Three-Way Catalytic Converter:

- Reduces CO, HC, NOx simultaneously
- Requires precise air-fuel ratio (14.7:1)

- Oxygen sensor feedback critical
- Operating temperature: 400-800°C

EGR (Exhaust Gas Recirculation):

- **Purpose:** Reduce NOx formation
- **How it works:**
  - Recirculates exhaust into intake
  - Lowers combustion temperature
  - Less NOx formed at lower temps
- **Components:**
  - EGR valve (controls flow)
  - EGR cooler (cools exhaust before mixing)
  - Vacuum or electronic control
- **Common issues:**
  - Carbon buildup in valve
  - Stuck open = rough idle, poor performance
  - Stuck closed = increased NOx, check engine light

EVAP System (Evaporative Emission Control):

- **Purpose:** Capture fuel vapors from tank
- **Components:**
  - Charcoal canister (stores vapors)
  - Purge valve (releases vapors to engine)
  - Vent valve
  - Fuel tank pressure sensor
  - Lines and hoses
- **Operation:**
  - Tank sealed, vapors stored in canister
  - Engine running, purge valve opens
  - Vapors burned in engine
- **Common issues:**
  - Loose fuel cap (most common!)
  - Cracked hoses
  - Failed purge valve
  - Check engine light codes

PCV (Positive Crankcase Ventilation):

- **Purpose:** Recirculate crankcase fumes
- **Why needed:**

    - Blow-by gases (combustion gases past rings)
    - Oil vapors
    - Would cause pressure buildup
- **How it works:**
    - One-way valve from crankcase to intake
    - Engine vacuum pulls fumes out
    - Burns them in combustion
- **Common issues:**
    - Clogged PCV valve
    - Oil consumption increase
    - Rough idle
    - Oil leaks from pressure buildup

Secondary Air Injection (Some Vehicles):

- Pumps fresh air into exhaust during cold start
- Helps catalyst warm up faster
- Less common on modern vehicles

Diesel Specific Systems:

- **DPF (Diesel Particulate Filter):**
    - Traps soot particles
    - Requires regeneration (burning off soot)
    - Can clog if used only for short trips
- **SCR (Selective Catalytic Reduction):**
    - Uses DEF/AdBlue (urea solution)
    - Reduces NOx
    - Separate fluid tank needs refilling
    - Common on commercial vehicles

Hour 3 (11:00 AM - 12:00 PM): OBD-II & Emission Monitoring

On-Board Diagnostics (OBD-II):

- **Purpose:** Monitor emission control systems
- **Required since:** BS4 (2017)
- **What it monitors:**
    - Catalytic converter efficiency
    - Oxygen sensor function
    - Evaporative system leaks

  - EGR operation
  - Misfires
  - All emission-related components

Check Engine Light (MIL - Malfunction Indicator Lamp):

- Yellow/amber engine symbol
- Indicates emission system problem
- **Solid light:** Problem detected, vehicle drivable
- **Flashing light:** Severe misfire, stop driving immediately
- Light stays on until problem fixed and codes cleared

Readiness Monitors:

- OBD system runs self-tests
- Must complete before emission testing
- Drive cycle needed after repairs
- "Not Ready" = haven't run tests yet

Emission Testing (PUC - Pollution Under Control):

- Required every 6 months (some states)
- Tests actual tailpipe emissions
- Must pass to register vehicle
- Checks: CO, HC, CO2, O2 levels
- Diesel: smoke opacity test

Technician Responsibility:

- Never disconnect emission controls
- Never suggest defeating systems
- Repair properly to meet standards
- Legal and ethical obligation
- Heavy fines for tampering

AFTERNOON SESSION: PRACTICAL (3 Hours)

Hour 1 (1:00 PM - 2:00 PM): Emission Component Identification

Activity 1: Complete Emission System Mapping (35 min)

- Identify on actual vehicle:
  - **Catalytic converter** (already know from Day 5)

    - **Oxygen sensors** (count: before/after catalyst)
    - **EGR valve location** (on intake or near manifold)
    - **EGR cooler** (if visible)
    - **Charcoal canister** (usually under vehicle near tank)
    - **Purge valve** (near intake manifold)
    - **PCV valve** (valve cover or crankcase)
    - **Fuel vapor lines** (small hoses from tank to canister)
    - **Fuel cap** (tethered or separate)
- Trace vapor flow path: tank → canister → purge valve → intake
- Draw emission system diagram
- Take photos of each component

Activity 2: Visual Emission Inspection (25 min)

- Check for common issues:
    - **Fuel cap:** Gasket damaged? Seals properly?
    - **EVAP hoses:** Cracks, disconnections?
    - **EGR valve:** Carbon buildup visible?
    - **PCV valve:** Disconnected? Clogged?
    - **Exhaust leaks:** Before catalyst (allows O2 in, false readings)
    - **Oxygen sensor wiring:** Damaged, melted?
- Create inspection checklist
- Document any issues found

Pass Criteria:

- ✅ Can locate catalytic converter and O2 sensors
- ✅ Can identify EGR valve
- ✅ Can locate EVAP canister and purge valve
- ✅ Can find PCV valve
- ✅ Creates complete emission system diagram

Hour 2 (2:00 PM - 3:00 PM): Basic Fluid Level Checking

Activity 1: Engine Oil Level Check (15 min)

- **Procedure:**
    1. Park on level surface
    2. Engine warm (been running) or cold (follow manual)
    3. Turn off engine, wait 5 minutes (oil drains to pan)
    4. Locate dipstick (usually yellow/orange handle)

5. Remove dipstick, wipe clean with rag
6. Reinsert fully
7. Remove and read level
8. Check against MIN/MAX marks
9. Wipe and reinsert dipstick

- **Interpreting level:**
  - Between MIN and MAX = OK
  - Below MIN = add oil
  - Above MAX = too full (can cause foaming)
  - Check color and smell
- **Oil condition inspection:**
  - **Normal:** Light brown, honey color
  - **Old:** Black (time to change)
  - **Milky/foamy:** Coolant mixing (bad!)
  - **Metal particles:** Engine wear (very bad!)
  - **Fuel smell:** Fuel dilution

Activity 2: Coolant Level Check (15 min)

- **SAFETY:** Only check when cold!
- **Procedure:**
  1. Locate coolant reservoir (translucent plastic tank)
  2. Check level marks: HOT/COLD or MIN/MAX
  3. Level should be at COLD mark when engine cold
  4. Check coolant color:
     - Green, pink, orange (depends on type)
     - Should be clean, not rusty
  5. If low, add correct coolant type
  6. Never open radiator cap when hot!
- **Coolant types (awareness):**
  - Green (traditional)
  - Orange (extended life)
  - Pink/red (different formula)
  - Don't mix types!

Activity 3: Brake Fluid Level Check (15 min)

- **Procedure:**
  1. Locate brake fluid reservoir (usually near firewall)
  2. Wipe top clean before opening
  3. Check level marks: MIN/MAX

4. Level should be near MAX
5. Check fluid color:
    - **New:** Clear or light amber
    - **Old:** Dark brown (moisture absorbed)
6. Never let reservoir go empty (air in system!)

- **Important notes:**
    - Brake fluid hygroscopic (absorbs moisture)
    - Level drops as brake pads wear (normal)
    - Sudden drop = leak (check brakes immediately!)
    - Don't overfill

Activity 4: Power Steering Fluid Check (15 min)

- **If vehicle has hydraulic power steering:**
    1. Locate reservoir (usually near engine, may have dipstick)
    2. Check when engine cold or hot (mark indicates)
    3. Read MIN/MAX marks
    4. Fluid color: Red or amber (ATF-based)
    5. Dark/burnt = needs changing
- **Note:** Many modern cars have electric power steering (no fluid!)

Pass Criteria:

- ✅ Can locate and check engine oil level correctly
- ✅ Can interpret oil condition
- ✅ Can check coolant level safely
- ✅ Can check brake fluid level
- ✅ Understands when levels indicate problems

Hour 3 (3:00 PM - 4:00 PM): Fluid Types & Specifications

Activity 1: Understanding Engine Oil Specifications (20 min)

- **Viscosity grades:**
    - SAE 5W-30, 10W-40, etc.
    - First number: Cold flow (W = Winter)
    - Second number: Hot viscosity
    - Lower first number = better cold start
    - Example: 5W-30 flows better than 10W-40 in cold
- **API ratings:**
    - SN, SP (petrol engines - S series)

    - CJ-4, CK-4 (diesel engines - C series)
    - Later letters = newer/better
    - Use manufacturer's recommended grade
- **Synthetic vs Mineral:**
    - Mineral (conventional): Basic oil
    - Semi-synthetic (blend): Better protection
    - Full synthetic: Best protection, longer intervals
    - Follow manufacturer recommendation
- **Reading oil containers:**
    - Locate viscosity grade
    - Find API rating
    - Check expiration date
    - Verify quantity (liters/quarts)

Activity 2: Coolant Types & Mixing (20 min)

- **Coolant = Antifreeze + Water**
    - Pure antifreeze = wrong (doesn't cool well)
    - Typical mix: 50/50 coolant and water
    - Some pre-mixed available
- **Types:**
    - **IAT (Green):** Traditional, 2-3 year life
    - **OAT (Orange/Red):** Extended life, 5 years
    - **HOAT (Various colors):** Hybrid, 5 years
    - **Don't mix different types!**
- **Why color matters:**
    - Color indicates chemistry
    - Wrong mix = sludge, corrosion
    - Always use manufacturer's specified type
- **Concentration:**
    - Too weak = poor freeze protection
    - Too strong = poor heat transfer
    - 50/50 mix standard for most climates

Activity 3: Brake Fluid Types (20 min)

- **DOT Ratings:**
    - **DOT 3:** Most common, lower boiling point
    - **DOT 4:** Higher boiling point, common in modern cars
    - **DOT 5:** Silicone-based, not compatible (rare)
    - **DOT 5.1:** High performance, compatible with DOT 3/4

    - Higher number = higher boiling point
- **Why boiling point matters:**
    - Braking creates heat
    - Fluid boils = vapor in lines = spongy pedal
    - Moisture absorption lowers boiling point
- **Compatibility:**
    - DOT 3 and DOT 4 can mix (not ideal)
    - DOT 5 CANNOT mix with others
    - Always use recommended type
- **Handling:**
    - Harmful to paint (clean spills immediately!)
    - Keep container sealed (absorbs moisture)
    - Never use old opened fluid

Practice:

- Students examine actual fluid containers
- Read labels for specifications
- Identify viscosity grades
- Check compatibility information
- Understand when to use which fluid

Pass Criteria:

- ✅ Can read engine oil viscosity grade
- ✅ Understands API ratings
- ✅ Can identify coolant types by color
- ✅ Knows brake fluid DOT ratings
- ✅ Understands importance of correct fluids

DAY 7 ASSESSMENT:

Practical Skills Checklist:

- [ ] Identifies emission control components
- [ ] Checks engine oil level correctly
- [ ] Interprets oil condition
- [ ] Checks coolant level safely
- [ ] Checks brake fluid level
- [ ] Reads fluid specifications from containers

- [ ] Understands fluid compatibility

Knowledge Check (Verbal):

- Name three harmful emissions and their effects
- What does the catalytic converter do?
- What does EGR stand for and what does it reduce?
- Which oil level is too high - above MAX or below MIN?
- Why shouldn't you mix different coolant types?

DAY 8: COOLING SYSTEM + COOLANT SERVICE BASICS

Learning Objectives:

- Understand heat transfer principles and engine cooling
- Identify all cooling system components
- Perform cooling system pressure testing
- Understand coolant service intervals

MORNING SESSION: THEORY (3 Hours)

Hour 1 (9:00 AM - 10:00 AM): Heat Transfer & Cooling Principles

Why Engines Need Cooling:

- Combustion creates ~2000°C temperatures
- Only 25-30% energy becomes mechanical work
- 30-35% becomes heat in coolant
- 30-35% exits in exhaust
- Must remove heat or engine melts/seizes

Optimal Operating Temperature:

- Typical: 85-95°C (185-205°F)
- Too cold: Poor efficiency, wear, emissions
- Too hot: Detonation, head gasket failure, warping
- Thermostat maintains optimal temp

Heat Transfer Methods:

- **Conduction:** Heat through solid materials

    - Combustion → cylinder walls → coolant
- **Convection:** Heat through fluid movement
    - Coolant circulation
    - Air flow through radiator
- **Radiation:** Heat through electromagnetic waves
    - Less significant in engines

Cooling System Design Philosophy:

- Liquid cooling (most common)
- Air cooling (older motorcycles, some older cars)
- Liquid cooling advantages:
    - More effective
    - More consistent
    - Quieter operation
    - Allows cabin heating

Hour 2 (10:00 AM - 11:00 AM): Cooling System Components

Radiator:

- **Function:** Heat exchanger (coolant to air)
- **Construction:**
    - Inlet tank (hot coolant from engine)
    - Outlet tank (cooled coolant returns)
    - Core (tubes and fins)
    - Tubes carry coolant
    - Fins increase surface area for airflow
- **Materials:**
    - Copper/brass (traditional, better heat transfer)
    - Aluminum/plastic (modern, lighter, cheaper)
- **Types:**
    - Cross-flow (side tanks)
    - Down-flow (top/bottom tanks)
- **Location:** Front of vehicle for maximum airflow

Radiator Cap (Pressure Cap):

- **Critical component!**
- Pressurizes system (typically 0.9-1.1 bar / 13-16 PSI)
- **Why pressure?** Raises boiling point

    - Water boils at 100°C at atmospheric pressure
    - At 1 bar pressure, boils at ~120°C
    - Prevents boiling, allows hotter operation
- **Two valves:**
    - Pressure relief valve (opens ~1 bar)
    - Vacuum valve (prevents collapse on cooling)
- **NEVER remove when hot!** Pressure release = violent boiling

Water Pump:

- **Function:** Circulate coolant through system
- **Types:**
    - Mechanical (belt-driven, most common)
    - Electric (some modern cars)
- **Location:** Front of engine, driven by serpentine belt
- **Construction:**
    - Impeller (spinning fins)
    - Sealed bearing
    - Shaft seal (prevents leaks)
- **Common failures:**
    - Bearing wear (noise, wobble)
    - Seal leak (coolant drips)
    - Impeller corrosion

Thermostat:

- **Function:** Control coolant flow to maintain temperature
- **Location:** Between engine and radiator (usually at engine outlet)
- **Operation:**
    - **Cold engine:** Closed, coolant circulates in engine only (bypass)
    - **Warm engine:** Opens gradually as temp rises
    - **Hot engine:** Fully open, coolant flows through radiator
- **Types:**
    - Wax pellet (expands with heat) - most common
    - Electronically controlled (newer cars)
- **Opening temperature:** Usually stamped on thermostat (82°C, 88°C, etc.)
- **Common failures:**
    - Stuck closed: Overheating
    - Stuck open: Never reaches operating temp, poor heater

Cooling Fans:

- **Function:** Force air through radiator when vehicle not moving
- **Types:**
  - **Mechanical (fan clutch):**
    - Belt-driven
    - Clutch engages when hot
    - Always spinning slightly
  - **Electric:**
    - Motor-driven, controlled by ECU
    - ON/OFF based on coolant temp sensor
    - More common now
    - Can run after engine off
- **When fan runs:**
  - Low speed driving
  - Idling
  - A/C on (high load)
  - Hot ambient temperature

Coolant Hoses:

- **Upper radiator hose:** Hot coolant from engine to radiator
- **Lower radiator hose:** Cooled coolant from radiator to engine
- **Heater hoses:** To/from cabin heater core
- **Bypass hose:** Allows circulation when thermostat closed
- **Materials:**
  - Reinforced rubber (EPDM)
  - Can have wire/fabric reinforcement
- **Common failures:**
  - Cracking (age, heat cycles)
  - Bulging (weak spots)
  - Collapsing (vacuum)

Heater Core:

- Mini radiator inside dashboard
- Hot coolant provides cabin heat
- Blower fan blows air through it
- If clogged: No heat in cabin
- If leaking: Coolant on floor, foggy windows

Coolant Temperature Sensor:

- Measures coolant temperature

- Signals ECU and dashboard gauge
- Critical for:
  - Fan operation
  - Fuel mixture adjustment when cold
  - Check engine light if overheating

Overflow/Expansion Tank:

- Captures expanding coolant when hot
- Returns it when cooling
- Allows checking level without opening radiator cap
- Translucent so you can see level

Hour 3 (11:00 AM - 12:00 PM): Cooling System Problems

Overheating Causes:

- **Low coolant level:** Leaks, never refilled
- **Stuck thermostat (closed):** No flow to radiator
- **Clogged radiator:** Mineral deposits, corrosion
- **Failed water pump:** Not circulating
- **Failed fan/fan clutch:** No airflow at idle
- **Blown head gasket:** Combustion gases in coolant
- **Radiator blockage:** Debris, bugs, dirt
- **Collapsed hose:** Restricts flow

Overheating Symptoms:

- Temperature gauge in red zone
- Steam from hood
- Sweet smell (coolant)
- Engine loses power
- Rough idle
- Check engine light

Never Reaching Temperature:

- **Stuck thermostat (open):** Always flowing to radiator
- **Symptoms:**
  - Gauge stays in cold zone
  - Poor heater performance

    - Poor fuel economy
    - Increased emissions

Coolant Leaks:

- **External leaks (visible):**
    - Hose connections
    - Radiator core
    - Water pump weep hole
    - Heater core (inside car)
    - Freeze plugs (core plugs)
- **Internal leaks (not visible):**
    - Head gasket failure
    - Cracked cylinder head/block
    - Symptoms:
        - White smoke from exhaust
        - Coolant loss with no visible leak
        - Oil looks milky (coolant mixing)
        - Bubbles in radiator when running

Coolant Contamination:

- **Oil in coolant:** Brown/muddy appearance
    - Head gasket leak
    - Oil cooler failure
    - Transmission cooler failure (if in radiator)
- **Rust in coolant:** Orange/brown deposits
    - Old coolant, no inhibitors left
    - Corroding internals
- **Electrolysis:** Electrical current in coolant
    - Eats metal components
    - Caused by grounding issues

Air in System:

- Prevents proper circulation
- Creates hot spots
- Must be bled out after service
- Bleeder valves or self-bleeding systems

AFTERNOON SESSION: PRACTICAL (3 Hours)

Hour 1 (1:00 PM - 2:00 PM): Cooling System Component Identification

Activity 1: Complete Cooling System Tracing (30 min)

- Identify all components on vehicle:
  - **Radiator:** Front of vehicle
  - **Radiator cap:** On radiator or on overflow tank
  - **Upper radiator hose:** From engine to radiator top
  - **Lower radiator hose:** From radiator bottom to engine
  - **Water pump:** Front of engine (belt-driven)
  - **Thermostat housing:** Usually where upper hose connects
  - **Cooling fan:** Behind radiator (mechanical or electric)
  - **Fan shroud:** Guides airflow
  - **Overflow tank:** Small plastic tank to side
  - **Heater hoses:** Two small hoses to firewall
  - **Coolant temperature sensor:** In thermostat housing or head
- Trace coolant flow path:
  - Engine → upper hose → radiator → lower hose → water pump → engine
- Draw complete cooling system diagram
- Note type of fan (mechanical/electric)

Activity 2: Visual Cooling System Inspection (30 min)

- **Coolant level check:** (already practiced Day 7)
  - Level at COLD mark when cold
  - Level rises when hot (expansion)
- **Hose inspection:**
  - Squeeze hoses (should be firm, not soft/squishy)
  - Look for cracks (especially at bends)
  - Check clamps for tightness
  - Look for bulges (weak spots)
- **Radiator inspection:**
  - Fins bent or damaged (airflow restriction)
  - Debris/bugs blocking fins
  - Look for green/pink stains (leaks)
  - Check mounting secure
- **Belt inspection:**
  - Condition (cracks, fraying)
  - Tension (press with thumb, ~1cm deflection)
  - Water pump pulley wobble

- **Leak evidence:**
  - Stains under vehicle (pink/green)
  - Wet spots on components
  - Crusty deposits (dried coolant)
  - Sweet smell

Pass Criteria:

- ✅ Can trace complete coolant flow path
- ✅ Can identify all major cooling components
- ✅ Can inspect hoses for condition
- ✅ Can check radiator for damage/blockage
- ✅ Documents findings properly

Hour 2 (2:00 PM - 3:00 PM): Cooling System Pressure Testing

Activity 1: Understanding Pressure Testing (15 min)

- **Why pressure test?**
  - Finds leaks before they cause problems
  - Simulates operating pressure
  - Verifies radiator cap function
  - Tests system integrity
- **Pressure tester components:**
  - Hand pump
  - Pressure gauge (0-2 bar range)
  - Adapters for different caps/necks
  - Universal adapter for overflow tanks
- **What we're checking:**
  - System holds pressure (no leaks)
  - Pressure drops = leak somewhere
  - Cap releases at correct pressure

Activity 2: Pressure Test Demonstration (20 min)

- Instructor demonstrates on vehicle:
  1. **Preparation:**
     - Engine MUST be cold
     - Remove radiator cap
     - Check coolant level (top up if needed)
  2. **Connect tester:**

    - Select correct adapter
    - Thread or clamp onto radiator neck
    - Ensure tight seal
3. **Pump to pressure:**
    - Pump to system rating (usually 0.9-1.1 bar)
    - Watch gauge
    - Note: Most caps rated 13-16 PSI / 0.9-1.1 bar
4. **Observe:**
    - Pressure should hold steady for 2 minutes
    - If drops: leak present
    - Walk around vehicle looking for leaks
    - Check hoses, radiator, water pump
    - Check ground under vehicle
5. **Test radiator cap:**
    - Install cap on tester
    - Pump to pressure
    - Should release at cap rating
    - Should hold below release pressure

Activity 3: Student Practice (Supervised) (25 min)

- Students practice pressure testing in pairs
- One operates tester, one observes/documents
- Switch roles
- Steps:
    - Check coolant level
    - Install tester correctly
    - Pump to specified pressure
    - Monitor gauge for 2 minutes
    - Record pressure reading every 30 seconds
    - Look for visible leaks
    - Test cap function
    - Remove tester carefully
    - Reinstall cap

Interpreting Results:

- **Holds pressure:** System good
- **Slow drop:** Small leak (investigate)
- **Quick drop:** Major leak (fix immediately)
- **Cap doesn't release at spec:** Replace cap

- **Cap doesn't seal:** Replace cap

Pass Criteria:

- ✅ Can explain purpose of pressure testing
- ✅ Can connect pressure tester correctly
- ✅ Can perform pressure test safely
- ✅ Can interpret test results
- ✅ Can test radiator cap function

Hour 3 (3:00 PM - 4:00 PM): Coolant Service Understanding

Activity 1: Coolant Condition Inspection (20 min)

- **Visual inspection of coolant:**
  - Remove coolant reservoir cap (cold only!)
  - Observe color and clarity:
    - **Good:** Bright, clear color (green, pink, orange)
    - **Bad:** Rusty brown, muddy, oil floating
  - Shine light through sample:
    - Should be translucent
    - Opaque = contaminated
- **Coolant testing strips (if available):**
  - Dip strip in coolant
  - Compare colors to chart
  - Tests:
    - pH level (should be ~7-11)
    - Freeze point protection
    - Inhibitor reserve (corrosion protection)
  - Low inhibitors = time to change
- **Hands-on practice:**
  - Students test coolant from multiple vehicles
  - Record condition (good/marginal/bad)
  - Determine: service needed?
  - Create service recommendations

Activity 2: Understanding Coolant Service (20 min)

- **When to service coolant:**
  - **Traditional (green):** Every 2-3 years or 40,000 km
  - **Extended life (orange):** Every 5 years or 100,000 km

    - **Or when testing shows degradation**
    - Follow manufacturer intervals
- **Why coolant degrades:**
    - Inhibitors deplete (corrosion protection)
    - pH changes (becomes acidic)
    - Contamination (rust, oil, dirt)
    - Loses freeze/boil protection
- **Coolant service procedure overview:**
    - **Drain:** Remove drain plug or lower hose
    - **Flush:** Run water through system
    - **Refill:** Correct coolant mix (50/50)
    - **Bleed air:** Remove trapped air
    - **Test:** Check for leaks, proper operation

NOTE: Students NOT performing drain/fill today (just awareness)

Activity 3: Temperature Gauge Understanding (20 min)

- **Dashboard temperature gauge:**
    - Usually center = normal (85-95°C)
    - Too far left = not warmed up yet
    - Too far right = overheating!
    - Some vehicles just have warning light
- **What to do if overheating:**
    1. Turn off A/C (reduces load)
    2. Turn on heater full blast (pulls heat from engine)
    3. Pull over safely
    4. Turn off engine
    5. Let cool (30+ minutes)
    6. Check coolant level (when cool!)
    7. Call for help if needed
    8. **NEVER remove cap when hot!**
- **Practice scenarios:**
    - Instructor describes situations
    - Students explain correct response:
        - "Gauge in red zone, steam from hood"
        - "Gauge barely moves, heater cold"
        - "Coolant on garage floor"
        - "Sweet smell in cabin, windows fogging"

Pass Criteria:

- ✅ Can assess coolant condition visually
- ✅ Understands service intervals
- ✅ Knows proper response to overheating
- ✅ Can interpret temperature gauge readings

DAY 8 ASSESSMENT:

Practical Skills Checklist:

- [ ] Identifies all cooling system components
- [ ] Traces coolant flow path correctly
- [ ] Inspects hoses and radiator
- [ ] Performs cooling system pressure test
- [ ] Tests radiator cap function
- [ ] Assesses coolant condition
- [ ] Interprets test results correctly

Knowledge Check (Verbal):

- Why do we pressurize the cooling system?
- What does the thermostat do?
- Name three causes of overheating
- What temperature is normal operating range?
- Why must the system be cold before opening the cap?

DAY 9: LUBRICATION SYSTEM + OIL SERVICE BASICS

Learning Objectives:

- Understand friction, wear, and lubrication principles
- Identify lubrication system components
- Understand oil properties and specifications
- Perform basic oil service procedures

MORNING SESSION: THEORY (3 Hours)

Hour 1 (9:00 AM - 10:00 AM): Friction & Lubrication Science

Understanding Friction:

- **Definition:** Resistance between moving surfaces
- **Generates heat and wear**
- **Types of friction:**
  - **Dry friction:** Metal-on-metal (very bad!)
  - **Boundary friction:** Thin oil film (bearings at startup)
  - **Hydrodynamic:** Full oil film separation (normal running)
- **Goal:** Achieve hydrodynamic lubrication

How Oil Reduces Friction:

- Creates film between surfaces
- Prevents metal-to-metal contact
- Film thickness critical:
  - Too thin: Metal contact, wear
  - Too thick: Power loss (churning)
- Oil pump maintains pressure to ensure film

Oil Functions (Multiple Jobs):

1. **Lubrication:** Reduce friction
2. **Cooling:** Carries heat away (30% of engine cooling!)
3. **Cleaning:** Suspends dirt and combustion byproducts
4. **Sealing:** Helps piston rings seal compression
5. **Corrosion prevention:** Protects metal surfaces
6. **Hydraulic medium:** Powers variable valve timing

Engine Wear:

- **Normal wear:** Microscopic, gradual over thousands of hours
- **Accelerated wear causes:**
  - Low oil level (starvation)
  - Wrong oil viscosity
  - Contaminated oil
  - Overheating
  - Cold starts (no oil film yet)
- **Wear patterns indicate problems:**
  - Scoring: Abrasive particles
  - Galling: Overheating, oil breakdown
  - Pitting: Corrosion

Hour 2 (10:00 AM - 11:00 AM): Lubrication System Components

Oil Pan/Sump:

- **Function:** Oil reservoir at bottom of engine
- **Capacity:** Typically 3-6 liters (varies by engine)
- **Features:**
  - Drain plug at lowest point
  - Baffles prevent oil sloshing
  - Some have cooling fins
- **Oil pickup tube:**
  - Submerged in oil
  - Screened opening (prevents large debris)
  - Feeds oil pump

Oil Pump:

- **Function:** Create pressure, circulate oil
- **Types:**
  - **Gear pump:** Two gears mesh, trap oil (most common)
  - **Rotor pump:** Inner/outer rotors create chambers
- **Drive method:**
  - Crankshaft driven (directly or via chain/belt)
  - Runs whenever engine runs
- **Pressure:**
  - Typical idle: 1-2 bar (15-30 PSI)
  - Typical running: 3-5 bar (45-75 PSI)
  - Varies with RPM and temperature
- **Pressure relief valve:**
  - Limits maximum pressure
  - Protects system from over-pressure

Oil Filter:

- **Function:** Remove contaminants from oil
- **Types:**
  - **Spin-on:** Complete unit, replace entire filter
  - **Cartridge:** Paper element only, housing stays
- **Filter media:**
  - Paper pleats (most common)
  - High surface area for dirt capacity
  - Traps particles down to 25-40 microns

- **Bypass valve:**
  - If filter clogs, allows oil to bypass
  - Better to have dirty oil than no oil!
- **Anti-drainback valve:**
  - Keeps oil in filter when engine off
  - Prevents dry start

Oil Pressure Sensor:

- **Function:** Monitor oil pressure
- **Types:**
  - Simple switch (light on = low pressure)
  - Variable sensor (gauge on dashboard)
- **Warning light:**
  - Comes on if pressure drops below safe level
  - **Never ignore oil light!** Pull over immediately

Oil Passages:

- **Galleries:** Drilled passages in block/head
- **Main gallery:** Feeds all lubrication points
- **Branch passages:** To specific bearings, camshaft
- **Spray jets:** Direct oil onto cylinder walls, pistons
- Oil flows through entire engine, returns to pan by gravity

Crankshaft Lubrication:

- **Main bearings:** Support crankshaft in block
  - Pressurized oil film
  - Oil fed through drilled passages
- **Rod bearings:** Connect rods to crankshaft
  - Oil fed through drilled holes in crankshaft
  - Oil throws outward from centrifugal force
- **Bearing clearances:** Typically 0.025-0.075mm
  - Too tight: Oil film breaks down, metal contact
  - Too loose: Low pressure, noisy

Camshaft & Valve Train Lubrication:

- Oil sprayed or fed to cam lobes
- Flows down to lifters/tappets
- Rocker arms receive oil through pushrods or passages
- Some systems use oil squirters

- Critical for cam lobe life (high pressure contact)

Piston & Cylinder Wall Lubrication:

- Oil scraped up by oil control rings
- Thin film on cylinder walls
- Some engines have piston cooling jets (spray oil on piston bottom)
- Critical during high load/high temperature

PCV System Integration:

- Recirculates crankcase vapors
- Prevents pressure buildup
- Covered in Day 7 emissions

Hour 3 (11:00 AM - 12:00 PM): Oil Properties & Selection

Viscosity (Most Important Property):

- **Definition:** Oil's resistance to flow (thickness)
- **Multi-grade oils:** 5W-30, 10W-40, etc.
- **Understanding grades:**
  - First number + W: Cold viscosity
    - "W" = Winter rating
    - Lower = flows better when cold
    - 0W, 5W, 10W, 15W, 20W
  - Second number: Hot viscosity (100°C)
    - Higher = thicker when hot
    - 20, 30, 40, 50, 60
- **Example: 5W-30**
  - Flows like 5W oil when cold (easy starting)
  - Protects like 30-weight oil when hot

Why Viscosity Matters:

- **Too thin (low viscosity):**
  - Good cold flow
  - Poor hot protection
  - Oil film breaks down
  - Increased wear
- **Too thick (high viscosity):**

    - Poor cold flow
    - Hard starting
    - Power loss (pumping resistance)
    - Reduced fuel economy
- **Must match manufacturer specification!**

Viscosity Index:

- How much viscosity changes with temperature
- Higher VI = less change (more stable)
- Synthetic oils have higher VI

Oil Types:

1. Mineral (Conventional):

- Refined from crude oil
- Basic protection
- Requires frequent changes (3,000-5,000 km)
- Cheapest option
- Good for older engines

2. Semi-Synthetic (Blend):

- Mix of mineral and synthetic
- Better protection than mineral
- Moderate cost
- Good middle ground
- Popular choice

3. Full Synthetic:

- Chemically engineered
- Best protection
- Longer change intervals (7,500-10,000 km)
- Better at temperature extremes
- Best for modern engines, turbo engines
- Most expensive
- Required by some manufacturers

API Service Ratings:

- **S = Petrol (Spark ignition):**
    - SA, SB, SC... SN, SP (current)

    - Later letters = newer/better
    - Current: SN (2010+), SP (2020+)
- **C = Diesel (Compression ignition):**
    - CA, CB, CC... CK-4 (current)
    - Commercial/heavy-duty ratings
- Always use rating specified by manufacturer

ACEA Ratings (European):

- Alternative to API
- A = Petrol, B = Diesel, C = Catalyst compatible
- Numbers indicate performance level

Manufacturer Approvals:

- VW 502.00, BMW LL-01, etc.
- Specific requirements beyond API/ACEA
- Critical for warranty compliance

Additives in Oil:

- **Detergents:** Keep surfaces clean
- **Dispersants:** Suspend contaminants
- **Anti-wear (ZDDP):** Protect under pressure
- **Viscosity index improvers:** Stabilize viscosity
- **Anti-oxidants:** Prevent oil breakdown
- **Friction modifiers:** Reduce friction further
- **Pour point depressants:** Improve cold flow
- Balanced additive package critical

Oil Change Intervals:

- **Manufacturer recommendation:** Follow it!
- **Severe service:** Shorter intervals
    - Short trips (engine doesn't fully warm)
    - Dusty conditions
    - Towing/heavy loads
    - Extreme temperatures
- **Normal service:** Standard intervals
- **Extended drains:** Only with correct oil and filter

Why Oil Changes Are Critical:

- Oil degrades over time and use

- Additives deplete
- Contamination builds up (soot, fuel, moisture)
- Oxidation (thickens oil)
- Acidic compounds form (corrosion)
- Never skip oil changes!

AFTERNOON SESSION: PRACTICAL (3 Hours)

Hour 1 (1:00 PM - 2:00 PM): Lubrication System Identification

Activity 1: Oil System Component Location (30 min)

- Identify on actual vehicle:
  - **Oil pan:** Bottom of engine (underneath)
  - **Drain plug:** Lowest point of pan (usually has washer)
  - **Oil filter:** Side of engine (spin-on or housing)
  - **Oil filler cap:** Top of engine (valve cover)
  - **Dipstick:** Side of engine (colored handle)
  - **Oil pressure sensor:** Near oil filter or on block
  - **Oil passages:** (not visible, but understand location)
  - **PCV valve:** Valve cover (emission system connection)
- Draw lubrication system diagram:
  - Pan → pickup → pump → filter → galleries → bearings → return to pan
  - Show oil filler and dipstick locations
  - Mark drain plug location

Activity 2: Oil Filter Identification (30 min)

- **Spin-on filter inspection:**
  - Locate on engine
  - Note size and threading
  - Read label: filter number, specs
  - Check mounting position (vertical/horizontal)
  - Look for leak evidence (oil stains)
- **Or cartridge filter:**
  - Locate housing
  - Identify how to access (remove cap/cover)
  - Understand element vs housing
- **Understanding filter markings:**
  - Part number

    - Thread size
    - Micron rating
    - Bypass pressure rating
- **Anti-drainback valve location:**
    - Usually in filter
    - Prevents oil from draining back when engine off
    - Critical for preventing dry starts

Pass Criteria:

- ✅ Can locate oil pan and drain plug
- ✅ Can identify oil filter type and location
- ✅ Can locate oil filler and dipstick
- ✅ Can trace basic oil flow path
- ✅ Creates accurate system diagram

Hour 2 (2:00 PM - 3:00 PM): Oil Inspection & Selection Practice

Activity 1: Used Oil Inspection (20 min)

- Drain small sample of used oil into clear container
- Inspect characteristics: **Color:**
    - **New:** Light amber, honey-colored
    - **Used but OK:** Dark brown to black (normal)
    - **Milky/foamy:** Coolant contamination (BAD!)
    - **Very dark/thick:** Overdue for change

  Consistency:
    - Pour oil: Should flow smoothly
    - Thick/sludgy: Oxidized, overheated
    - Thin/watery: Fuel dilution

  Contaminants:
    - Smell for fuel (gasoline odor)
    - Feel for grit (metal particles)
    - Look for particles in light

  Blotter test (if available):
    - Drop oil on absorbent paper
    - Should form even ring
    - Soot ring in center normal
    - Crud/particles = contamination
- Create oil condition report

- Determine: OK to continue or change now?

Activity 2: Oil Specification Reading (20 min)

- Provide various oil containers
- Students practice reading labels: **Find and record:**
  - Viscosity grade (5W-30, 10W-40, etc.)
  - API rating (SN, SP, etc.)
  - Type (mineral, semi-synthetic, full synthetic)
  - Manufacturer approvals (if any)
  - Volume/quantity
  - Expiration date
- **Matching oil to vehicle:**
  - Check owner's manual specification
  - Find matching oil on shelf
  - Understand what each spec means
  - Calculate quantity needed (check capacity)

Activity 3: Filter Selection (20 min)

- Using filter catalog or online database:
  - Input vehicle make/model/year/engine
  - Identify correct filter part number
  - Cross-reference with available filters
  - Note OEM vs aftermarket options
  - Compare prices (quality vs cost)
- **Filter quality indicators:**
  - Filtration efficiency (percentage)
  - Dirt capacity (grams)
  - Construction quality
  - Brand reputation
- Understanding when premium filter justified:
  - Severe service conditions
  - Extended oil change intervals
  - High-performance engines

Pass Criteria:

- ✅ Can assess oil condition visually
- ✅ Can read oil specifications from container
- ✅ Can match oil spec to vehicle requirements
- ✅ Can identify correct filter for vehicle

Hour 3 (3:00 PM - 4:00 PM): Basic Oil Service Procedure (Observation & Understanding)

Activity 1: Oil Change Procedure Demonstration (25 min)

- Instructor demonstrates complete oil change
- Students observe and take notes

Preparation:

1. Warm engine (5 min run time) - hot oil drains faster
2. Lift vehicle safely
3. Locate drain plug and filter
4. Position drain pan under drain plug
5. Have new oil, filter, drain plug washer ready

Draining Oil:

1. Loosen drain plug slowly (last few turns by hand)
2. Let drain plug fall into pan (have rag ready)
3. Let oil drain completely (5-10 minutes)
4. Inspect drain plug and washer
5. Clean drain plug threads
6. Install new crush washer (if required)
7. Reinstall drain plug - proper torque (25-30 Nm typically)
    - Not too tight (strips threads)
    - Not too loose (leaks)

Filter Replacement:

- **Spin-on filter:**
    1. Position drain pan under filter
    2. Use filter wrench to loosen
    3. Unscrew by hand when loose
    4. Check old filter gasket came off
    5. Clean mounting surface on engine
    6. Apply thin film of new oil to new filter gasket
    7. Install new filter hand-tight
    8. Tighten 3/4 to 1 full turn after gasket contacts
    9. Don't over-tighten!
- **Cartridge filter:**
    1. Remove cover/cap

2. Remove old element
3. Clean housing
4. Install new O-rings (lube with oil)
5. Install new element
6. Reinstall cover with proper torque

Filling New Oil:

1. Lower vehicle (if lifted)
2. Remove oil filler cap
3. Place funnel in filler neck
4. Pour in specified amount (check manual)
   - Slightly less than capacity initially
5. Install filler cap
6. Start engine, let run 1 minute
7. Check for leaks at drain plug and filter
8. Turn off engine, wait 2 minutes
9. Check oil level with dipstick
10. Add more if needed to reach FULL mark
11. Recheck for leaks

Documentation:

1. Record mileage/date
2. Record oil type and quantity used
3. Record filter part number
4. Place sticker on door jamb or windshield
5. Update service records

NOTE: Students are OBSERVING today, not performing. Hands-on oil changes would be in specialization tracks.

Activity 2: Used Oil Disposal (15 min)

- **Environmental responsibility:**
  - Used oil is hazardous waste
  - Never pour down drain
  - Never dump on ground
  - Recycle properly
- **Proper disposal:**
  - Store in approved containers
  - Label clearly
  - Take to recycling center or auto parts store

    - Many locations accept free
    - Commercial shops have pickup service
- **Used filter disposal:**
    - Drain completely (24 hours)
    - Dispose as metal waste or with used oil
    - Some locations recycle separately

Activity 3: Service Interval Planning (20 min)

- **Calculating next service:**
    - Current mileage: _ km
    - Service interval: 5,000 km (example)
    - Next service at: _ km
    - Or time-based: 6 months
- **Service reminder systems:**
    - Dashboard indicator
    - "Change Oil Soon" light
    - Percentage-based systems
    - Reset procedure after service
- **Record keeping:**
    - Maintain service log
    - Helps resale value
    - Proves proper maintenance
    - Useful for diagnostics
- **Creating service plan:**
    - Student vehicles or example
    - When was last change?
    - What interval to use?
    - When is next due?
    - What oil specification needed?

Pass Criteria:

- ✅ Understands complete oil change procedure
- ✅ Knows proper drain plug torque importance
- ✅ Understands filter installation technique
- ✅ Knows proper oil level checking procedure
- ✅ Understands disposal requirements
- ✅ Can calculate service intervals

DAY 9 ASSESSMENT:

Practical Skills Checklist:

- [ ] Identifies all lubrication system components
- [ ] Can check oil level correctly (from Day 7)
- [ ] Can assess oil condition visually
- [ ] Reads oil specifications from containers
- [ ] Matches oil spec to vehicle requirements
- [ ] Selects correct filter for vehicle
- [ ] Understands oil change procedure sequence
- [ ] Knows proper disposal methods

Knowledge Check (Verbal):

- What are the five functions of engine oil?
- What does 5W-30 mean?
- Why is the first number important in cold climate?
- What does API SN mean?
- Why must we replace the drain plug washer?
- What happens if oil level is too low?

DAY 10: CLUTCH SYSTEM + CLUTCH BASICS

Learning Objectives:

- Understand clutch operation principles
- Identify clutch components
- Understand clutch adjustment basics
- Recognize clutch wear symptoms

MORNING SESSION: THEORY (3 Hours)

Hour 1 (9:00 AM - 10:00 AM): Clutch Fundamentals

Why We Need a Clutch:

- Engine always wants to run
- Wheels sometimes need to stop
- Need device to connect/disconnect engine from transmission

- Allows smooth engagement of power
- Enables gear changing without grinding

Clutch Location:

- Between engine and transmission
- Mounted to engine flywheel
- Transmission input shaft passes through center

Basic Operating Principle:

- **Clutch Pedal UP (engaged):**
  - Friction disc clamped between flywheel and pressure plate
  - Engine power transmitted to transmission
  - Like squeezing a piece of paper between hands
- **Clutch Pedal DOWN (disengaged):**
  - Pressure plate released
  - Friction disc free to spin independently
  - No power transmission

Friction Principle:

- Clutch works on friction between surfaces
- Coefficient of friction between disc and flywheel/pressure plate
- Smooth engagement requires gradual increase in friction
- Driver controls with pedal position

Hour 2 (10:00 AM - 11:00 AM): Clutch Components

Flywheel:

- **Functions:**
  - Smooth out engine pulses
  - Provides inertia for smooth running
  - First friction surface for clutch
  - Starter ring gear around outside edge
- **Types:**
  - Single-mass (solid, traditional)
  - Dual-mass (two sections with springs, smoother)
- **Surface:**
  - Must be flat and smooth

    - Heat cracks common on old flywheels
    - Can be resurfaced (machined) if worn
    - Balance critical (factory balanced)

Clutch Disc (Friction Disc):

- **Center hub:** Splined to transmission input shaft
- **Friction material:** Both sides, like brake pads
    - Organic (traditional)
    - Ceramic (performance, aggressive)
    - Kevlar (high performance)
- **Torsion springs:** Dampen shock loads
    - Inside hub
    - Allow slight rotation (cushioning)
- **Marcel spring (wave washer):** Smooth engagement
- **Wear:**
    - Friction material wears like brake pads
    - Eventually gets thin
    - Normal wear item (60,000-100,000 km typical)

Pressure Plate Assembly:

- **Components:**
    - Pressure plate (heavy metal plate)
    - Diaphragm spring (Belleville spring)
    - Cover (bolts to flywheel)
    - Release bearing contact points (fingers)
- **Function:**
    - Clamps disc to flywheel when engaged
    - Releases disc when pedal pressed
- **Types:**
    - Diaphragm spring (most common modern)
    - Coil spring (older, heavy-duty)
- **Clamping force:**
    - Typical: 2000-5000 N (200-500 kg force)
    - Heavy-duty: even higher
    - Determines torque capacity

Release Bearing (Throw-out Bearing):

- **Function:** Pushes on pressure plate fingers
- **Types:**

    - Mechanical (physically pushed by fork)
    - Hydraulic (concentric slave cylinder)
- **Rotation:**
    - Bearing must rotate when pressed
    - If seized: noise, failure
- **Wear:**
    - Eventually wears out
    - Replaced with clutch typically

Clutch Fork/Lever:

- Mechanical linkage from slave cylinder to release bearing
- Pivots on ball stud
- Pushes release bearing into pressure plate
- Can bend if misadjusted or abused

Pilot Bearing/Bushing:

- Supports tip of transmission input shaft
- Located in center of crankshaft/flywheel
- Bronze bushing or sealed bearing
- Failure: noise when clutch depressed

Hour 3 (11:00 AM - 12:00 PM): Clutch Hydraulic & Cable Systems

Cable-Operated Clutch (Older Vehicles):

- **Components:**
    - Clutch pedal
    - Cable (inner wire in outer sheath)
    - Clutch fork
    - Adjuster at transmission end
- **Operation:**
    - Pedal pulls cable
    - Cable pulls fork
    - Fork pushes release bearing
    - Mechanical advantage from lever ratio
- **Adjustment:**
    - Free play at pedal required (typically 2-3cm)
    - Adjusted at cable end
    - Too little: clutch slips (always partly disengaged)

    - Too much: hard to disengage completely
- **Maintenance:**
    - Cable can stretch over time
    - Cable can fray/break
    - Requires periodic lubrication

Hydraulic Clutch System (Modern):

- **Components:**
    1. **Clutch pedal:** With pushrod
    2. **Master cylinder:** Creates hydraulic pressure
        - Reservoir with brake fluid (DOT 3/4)
        - Piston and seal
        - Connected to pedal
    3. **Hydraulic line:** Steel or braided
    4. **Slave cylinder:** Converts pressure to motion
        - External: Pushes clutch fork
        - Concentric: Surrounds input shaft, pushes release bearing directly
    5. **Release bearing:** (integrated with concentric slave)

Hydraulic Operation:

- Pedal depressed → master cylinder piston moves
- Fluid pressure transmitted through line
- Slave cylinder piston extends
- Pushes fork or release bearing
- No adjustment needed (self-adjusting)

Hydraulic Advantages:

- Lighter pedal effort
- No adjustment needed
- More consistent feel
- Less maintenance
- Can route anywhere (not direct line like cable)

Hydraulic Issues:

- **Air in system:** Spongy pedal
    - Requires bleeding
- **Fluid leak:** Loss of pressure
    - Master or slave cylinder seal
    - Line damage

- **Fluid contamination:** Swollen seals
  - Wrong fluid type used

Clutch Pedal Free Play:

- **Cable system:** 2-3cm free play at pedal
- **Hydraulic system:** Self-adjusting (no external adjustment)
- **Too little free play:**
  - Release bearing always touching pressure plate
  - Premature bearing wear
  - Clutch may slip
- **Too much free play:**
  - Clutch doesn't fully disengage
  - Grinding when shifting
  - Hard to get into gear

AFTERNOON SESSION: PRACTICAL (3 Hours)

Hour 1 (1:00 PM - 2:00 PM): Clutch System Identification

Activity 1: External Component Location (30 min)

- Identify accessible clutch components:
  - **Clutch pedal:** In driver footwell
  - **Pedal free play:** Measure at top of pedal
  - **Master cylinder:** Firewall, driver side
    - Fluid reservoir (shared with brake or separate)
    - Check fluid level
  - **Hydraulic line:** From master to slave
  - **Slave cylinder:** On transmission bellhousing
    - External type (has pushrod)
    - Or concentric type (inside bellhousing, not visible)
  - **Clutch fork:** Sometimes visible through bellhousing inspection hole
  - **Bellhousing:** Large casting between engine and transmission
- Take photos and label each component
- Draw clutch hydraulic system diagram

Activity 2: Clutch Pedal Free Play Check (30 min)

- **Understanding free play:**
  - Distance pedal moves before resistance felt

- Must be present for cable systems
- Indicates adjustment is correct

Measurement procedure (Cable System):

1. Measure pedal height from floor (relaxed position)
2. Push pedal gently until resistance felt
3. Measure this distance
4. Difference = free play
5. Compare to specification (typically 20-30mm)

Hydraulic systems:

- Usually self-adjusting
- Free play not adjustable
- Check for proper pedal return

Clutch pedal operation check:

1. Press pedal slowly
2. Feel for smooth, consistent resistance
3. Listen for noise (squeaking, grinding)
4. Check pedal returns fully
5. Check for fluid leaks at master cylinder

Student practice:

- Each student checks clutch pedal on vehicle
- Measures free play (if cable)
- Records findings
- Compares to specifications

Pass Criteria:

- ✅ Can locate clutch hydraulic components
- ✅ Can measure clutch pedal free play
- ✅ Understands significance of free play
- ✅ Can check clutch pedal operation

Hour 2 (2:00 PM - 3:00 PM): Clutch Operation & Friction Point Practice

Activity 1: Understanding the Friction Point (20 min)

- **Friction point definition:**
  - Point where clutch begins to engage
  - Where power starts transmitting
  - Where vehicle begins to move (if in gear)

Finding the friction point:

- Instructor demonstrates in stationary vehicle:
  1. Engine running, transmission in neutral
  2. Press clutch pedal fully
  3. Select first gear
  4. Slowly release clutch pedal
  5. Feel for friction point (engine RPM drops slightly)
  6. Hold pedal at this point
  7. This is where engagement begins

Significance:

- **Too high in pedal travel:**
  - Clutch worn (thin disc)
  - May slip under load
- **Too low in pedal travel:**
  - Clutch new or over-adjusted
  - Normal
- **No clear friction point:**
  - Clutch problems
  - Hydraulic issues (air in system)

Student practice:

- Students locate friction point on vehicles
- Not actually driving, just finding engagement point
- Note position in pedal travel
- Feel engine load change

Activity 2: Clutch Fluid Check (20 min)

- **Locating reservoir:**
  - Sometimes shared with brake system
  - Sometimes separate reservoir
  - Near master cylinder on firewall

Checking procedure:

1. Wipe top clean (prevent contamination)
2. Check level against MIN/MAX marks
3. Check fluid color:
   - Should be clear or light amber
   - Dark = contaminated or old
4. Check for leaks around master cylinder
5. If low, top up with DOT 3 or DOT 4 (check cap marking)
6. Do NOT overfill

Fluid maintenance:

- Should be changed every 2-3 years
- Absorbs moisture (hygroscopic)
- Moisture lowers boiling point
- Can cause spongy pedal

Activity 3: Clutch Adjustment Overview (20 min)

- **Cable clutch adjustment:**
  - Locate adjuster nut at transmission end
  - Turn to lengthen or shorten cable
  - Measure pedal free play after each adjustment
  - Lock adjuster when correct

NOTE: Students NOT adjusting today (just understanding process)

Proper adjustment critical:

- Incorrect adjustment causes premature wear
- Too loose: Grinding, hard shifting
- Too tight: Slipping, burning

Pass Criteria:

- ✅ Can locate friction point
- ✅ Can check clutch fluid level
- ✅ Understands adjustment process
- ✅ Recognizes symptoms of mis-adjustment

Hour 3 (3:00 PM - 4:00 PM): Clutch Problems & Symptoms

Activity 1: Clutch Problem Diagnosis (Discussion) (30 min)

Common clutch problems and symptoms:

1. Slipping Clutch:

- **Symptoms:**
  - Engine RPM increases but vehicle doesn't accelerate proportionally
  - Burning smell (like brakes)
  - Loss of power on hills or acceleration
- **Causes:**
  - Worn friction disc (most common)
  - Oil contamination on disc
  - Weak pressure plate springs
  - Incorrect adjustment (cable too tight)
- **Test:** Accelerate hard in high gear; if RPM rises without speed increasing = slipping

2. Clutch Won't Disengage (Dragging):

- **Symptoms:**
  - Hard to shift gears
  - Grinding when selecting gear
  - Vehicle creeps forward in gear with clutch depressed
- **Causes:**
  - Incorrect adjustment (too much free play)
  - Air in hydraulic system
  - Warped disc
  - Failed release bearing
  - Broken clutch fork

3. Clutch Chatter/Judder:

- **Symptoms:**
  - Vibration when releasing clutch
  - Grabby engagement
  - Jerky starts
- **Causes:**
  - Contaminated friction disc (oil, grease)
  - Worn pressure plate
  - Warped flywheel
  - Worn engine/transmission mounts

4. Noises:

- **Squealing when depressing pedal:**

    - Failed release bearing
    - Or pilot bearing if noise when clutch released
- **Grinding when shifting:**
    - Clutch not fully disengaging
    - Or transmission synchronizer problem
- **Rattling at idle (pedal up):**
    - Worn torsion springs in disc
    - Or dual-mass flywheel failure

5. Heavy/Hard Pedal:

- **Causes:**
    - Cable binding/needs lubrication
    - Hydraulic system problem
    - Pressure plate too heavy for application
    - Worn clutch fork pivot

6. Spongy Pedal:

- **Hydraulic systems only:**
    - Air in system
    - Leaking master/slave cylinder
    - Worn seals

Activity 2: Visual Inspection (When Possible) (15 min)

- If vehicle has inspection cover on bellhousing:
    - Remove cover
    - Inspect visible clutch components:
        - Friction disc edge (wear visible?)
        - Release bearing condition
        - Fork position and condition
    - Use flashlight and mirror
    - Look for oil leaks (indicates contamination)

Activity 3: Customer Communication Practice (15 min)

- **Scenario-based practice:**
    - Instructor describes customer complaint
    - Students identify likely problem
    - Students explain diagnosis in simple terms
    - Students explain repair needed and cost estimate

Example scenarios:

- "My car smells like burning when I drive up hills"
- "I can't get my car into first gear when stopped"
- "My car shudders when I start moving"
- "There's a squealing noise when I press the clutch pedal"

Professional communication:

- Avoid jargon with customers
- Explain cause and effect
- Provide cost estimate
- Explain consequences of delaying repair
- Give options (if available)

Pass Criteria:

- ✅ Can identify symptoms of common clutch problems
- ✅ Can explain likely causes
- ✅ Can perform basic visual inspection
- ✅ Can communicate findings to customers

DAY 10 ASSESSMENT:

Practical Skills Checklist:

- [ ] Identifies clutch system components
- [ ] Measures clutch pedal free play
- [ ] Checks clutch fluid level
- [ ] Locates clutch friction point
- [ ] Recognizes symptoms of clutch problems
- [ ] Explains problems in customer-friendly terms

Knowledge Check (Verbal):

- What is the clutch's primary function?
- Name the three main clutch components
- What causes a slipping clutch?
- What does excessive pedal free play cause?
- Why is hydraulic fluid level important?

DAY 11: TRANSMISSION SYSTEMS + BASIC CHECKS

Learning Objectives:

- Understand transmission purpose and types
- Identify different transmission systems
- Understand gear ratio concepts
- Perform basic transmission fluid checks

MORNING SESSION: THEORY (3 Hours)

Hour 1 (9:00 AM - 10:00 AM): Transmission Fundamentals

Why We Need a Transmission:

- Engine produces power in narrow RPM range (typically 1500-6000 RPM)
- Wheels need different speeds (0-200+ km/h)
- Need to multiply torque for acceleration
- Need to reduce engine speed for cruising
- **Solution:** Multiple gear ratios

Gear Ratio Concept:

- **Definition:** Relationship between input and output speed
- **Example:** 3:1 ratio
  - Input shaft turns 3 times
  - Output shaft turns 1 time
  - Torque multiplied by 3
  - Speed divided by 3

Understanding Gears:

- **First gear:** Highest ratio (3.5:1 to 4.5:1)
  - Maximum torque multiplication
  - Lowest speed
  - For starting and heavy loads
- **Second gear:** Medium-high ratio (2.5:1 to 3:1)
  - Less torque, more speed
- **Third gear:** Medium ratio (1.5:1 to 2:1)
- **Fourth gear:** Low ratio (1:1 or 1.2:1)
  - Direct drive or close to it
- **Fifth/Sixth gear:** Overdrive (0.7:1 to 0.9:1)

- Output faster than input
- Fuel economy at cruise
- Lower RPM, less noise

Reverse Gear:

- Uses idler gear to reverse rotation
- Typically high ratio (like 1st gear)
- Limited to low speeds

Hour 2 (10:00 AM - 11:00 AM): Manual Transmission

Manual Transmission Components:

1. Input Shaft:

- Splined to clutch disc
- Rotates at engine speed (when clutch engaged)
- Has gear cluster (constant mesh)

2. Output Shaft:

- Connects to driveshaft (RWD) or differential (FWD)
- Has gears that can be locked to shaft
- Delivers power to wheels

3. Countershaft (Layshaft):

- Always driven by input shaft
- Carries intermediate gears
- Transmits power to output shaft

4. Gears:

- **Constant mesh:** Always spinning together
- **Sliding mesh:** Older design, gears slide into engagement
- **Synchromesh:** Modern, synchronizes speeds before engagement

5. Synchronizers (Synchros):

- Brass or carbon friction cones
- Match gear speed to shaft speed before locking
- Allows smooth shifting without grinding
- Wear item (but long-lasting typically)

6. Shift Forks:

- Moved by shift linkage
- Slide synchronizer assemblies
- Select different gears

7. Shift Linkage:

- Connects shifter to gears inside transmission
- **Cable:** Modern, flexible routing
- **Mechanical rods:** Traditional, direct feel
- **Pattern:** H-pattern (most common)

Manual Transmission Operation:

- Neutral: No gears locked, output shaft free
- Selecting gear:
  1. Depress clutch (disconnect engine)
  2. Move shifter
  3. Shift fork moves synchronizer
  4. Synchronizer matches speeds
  5. Gear locks to output shaft
  6. Release clutch (connect engine)
  7. Power flows through selected gear ratio

Power Flow Example (3rd Gear):

- Engine → clutch → input shaft → 3rd gear on countershaft → 3rd gear on output shaft → driveshaft → wheels

Manual Transmission Advantages:

- More driver control
- Better fuel economy (traditionally)
- Less maintenance
- Lower cost
- More engaging to drive

Disadvantages:

- Requires skill/learning
- More effort in traffic
- Clutch wears (replacement cost)

Hour 3 (11:00 AM - 12:00 PM): Automatic Transmission Types

Torque Converter Automatic (Traditional):

Key Difference from Manual:

- No clutch pedal
- Uses torque converter instead of friction clutch
- Gear changes automatic based on speed/load

Torque Converter:

- Fluid coupling
- **Components:**
  - **Impeller:** Connected to engine, pumps fluid
  - **Turbine:** Connected to transmission, driven by fluid
  - **Stator:** Between them, redirects fluid flow
- **Function:**
  - Allows engine to run while vehicle stopped
  - Multiplies torque at low speeds
  - Becomes 1:1 coupling at cruise
- **Lock-up clutch:** Locks impeller to turbine at highway speeds (efficiency)

Automatic Transmission Internals:

- Planetary gear sets (multiple)
- Clutch packs (engage/disengage planetary members)
- Bands (wrap around drums, hold components)
- Valve body (hydraulic control)
- TCM (Transmission Control Module) - electronic brain
- Typically 4-8 speeds modern

Advantages:

- Easy to drive
- Smooth shifts
- No clutch wear
- Good for traffic

Disadvantages:

- More complex
- More expensive
- Lower fuel economy (traditional designs)

- Heavier

AMT (Automated Manual Transmission):

- Manual transmission with automated clutch and shifting
- Electronic/hydraulic actuators shift gears
- Driver just uses throttle and brake
- **Examples:** Maruti AGS, Tata AMT
- **Advantages:**
  - Fuel economy like manual
  - Easy to drive like automatic
  - Lower cost than full automatic
- **Disadvantages:**
  - Shift delays (noticeable gaps)
  - Jerky at low speeds
  - Clutch still wears (but automatically controlled)

CVT (Continuously Variable Transmission):

- No fixed gears
- Uses belt/chain between variable-diameter pulleys
- **Operation:**
  - Pulleys change diameter ratio smoothly
  - Infinite ratios within range
  - Keeps engine at optimal RPM
- **Advantages:**
  - Smoothest acceleration
  - Best fuel economy
  - Simple mechanically
- **Disadvantages:**
  - "Rubber band" feel (engine RPM disconnected from speed)
  - Belt/chain wear
  - Noise (whining under acceleration)
  - Limited torque capacity
- **Common in:** Honda, Nissan, Subaru

DCT/DSG (Dual-Clutch Transmission):

- Two separate manual transmissions
- Two clutches (one for even gears, one for odd gears)
- **Operation:**
  - Pre-selects next gear while in current gear

    - Clutch swap happens in milliseconds
    - Very fast shifts
- **Advantages:**
    - Fastest shifts (0.1 seconds)
    - Excellent fuel economy
    - Performance feel
- **Disadvantages:**
    - Expensive
    - Complex (more failure points)
    - Jerky at low speeds (some designs)
- **Common in:** VW/Audi (DSG), Hyundai, Kia

Comparison Summary:

Type	Gears	Efficiency	Cost	Smoothness	Complexity
Manual	Fixed (5-6)	High	Low	Driver skill	Simple
Automatic	Fixed (4-8)	Medium	High	Smooth	Complex
AMT	Fixed (5-6)	High	Medium	Jerky	Medium
CVT	Infinite	Highest	Medium	Very Smooth	Medium
DCT	Fixed (6-7)	High	Highest	Fast/Smooth	Most Complex

AFTERNOON SESSION: PRACTICAL (3 Hours)

Hour 1 (1:00 PM - 2:00 PM): Transmission Identification

Activity 1: Transmission Type Recognition (30 min)

- Identify transmission type on various vehicles:

External identification clues:

- **Manual:**
    - Clutch pedal present
    - Shift lever with gate pattern
    - Clutch master cylinder visible
    - Simpler appearance
- **Automatic (Torque Converter):**
    - No clutch pedal
    - P-R-N-D-S shift pattern

    - Larger transmission housing
    - Dipstick often present (fluid check)
- **AMT:**
    - No clutch pedal
    - May have +/- shift option
    - Badge (AGS, AMT, etc.)
    - Small actuator visible on transmission
- **CVT:**
    - No clutch pedal
    - May say "CVT" on shift pattern
    - Badge on vehicle
    - Transmission case unique shape
- **DCT:**
    - No clutch pedal
    - May have paddle shifters
    - Badge (DSG, DCT, PDK, etc.)
    - Twin clutch housings visible

Activity 2: Transmission Component Location (30 min)

- Identify transmission components on lifted vehicle:
    - **Transmission housing:** Behind engine
    - **Bell housing:** Front section (houses clutch)
    - **Shift linkage:** From shifter to transmission
    - **Transmission mount:** Rubber mount supporting weight
    - **Output shaft/flange:** Connects to driveshaft or axles
    - **Breather:** Small vent on top
    - **Drain plug:** Bottom of housing (if accessible)
    - **Fill plug:** Side of housing
    - **Speed sensors:** Usually on housing
    - **Dipstick:** (automatic only, not all models)
- Draw transmission location diagram
- Note relationship to engine and drivetrain

Pass Criteria:

- ✅ Can identify transmission type from external clues
- ✅ Can locate transmission on vehicle
- ✅ Can identify shift linkage
- ✅ Can find fluid check points

Hour 2 (2:00 PM - 3:00 PM): Transmission Fluid Check (Automatic)

Activity 1: Automatic Transmission Fluid (ATF) Check (25 min)

NOTE: Some modern automatics have NO dipstick (sealed for life or dealer-only check)

If dipstick present:

1. **Start engine, let warm up (5 min)**
2. **Park on level surface**
3. **Keep engine running** (important!)
4. **Shift through all gears** (P-R-N-D-2-1-D-N-P)
5. **Return to Park**
6. **Locate ATF dipstick** (usually red handle, rear of engine bay)
7. **Remove dipstick, wipe clean**
8. **Reinsert fully**
9. **Remove and read level**
   - Should be in "HOT" range
   - Between MIN and MAX marks
10. **Check fluid condition:**
    - **Color:** Should be red/pink (new) or brownish-red (used)
    - **Smell:** Should have slight petroleum smell
    - **Clarity:** Should be translucent, not opaque
11. **Replace dipstick fully**

Fluid condition assessment:

- **Good:** Red or light brown, transparent, normal smell
- **Serviceable:** Dark brown, still translucent
- **Bad:** Black, opaque, burnt smell (needs change)
- **Very bad:** Metallic particles, milky (internal damage)

Activity 2: Manual Transmission Fluid Check (20 min)

Manual transmissions typically NO dipstick:

- Must check at fill plug
- Vehicle must be level
- Requires lifting usually

Procedure (demonstration):

1. Lift vehicle safely
2. Locate fill plug (side of transmission, upper)

3. Remove fill plug
4. Fluid should be level with bottom of hole
5. If low, add fluid until level with hole
6. Reinstall fill plug with washer

Manual transmission fluids:

- Engine oil (some models)
- Gear oil (GL-4 or GL-5, 75W-90 typical)
- MTF (Manual Transmission Fluid - special formula)
- **Always check manufacturer specification!**

Activity 3: CVT Fluid Check (15 min)

- **CVT fluid unique - never substitute!**
- **Most CVTs sealed (no dipstick)**
- Requires dealer-level check often
- **If accessible:**
  - Similar to automatic procedure
  - Must be correct CVT fluid
  - Color typically green or clear

Fluid Service Intervals:

- **Manual:** 40,000-60,000 km or 3-4 years
- **Automatic:** 40,000-80,000 km (or "lifetime" sealed)
- **CVT:** 40,000-60,000 km (critical - never skip!)
- **DCT:** 40,000-60,000 km
- **Always follow manufacturer intervals!**

Pass Criteria:

- ✅ Can check automatic transmission fluid level correctly
- ✅ Can assess ATF condition
- ✅ Understands manual transmission check procedure
- ✅ Knows different fluid types exist

Hour 3 (3:00 PM - 4:00 PM): Basic Transmission Problem Recognition

Activity 1: Common Transmission Problems (Discussion) (25 min)

Manual Transmission Problems:

1. Hard to Shift:

- **Causes:**
  - Clutch not fully disengaging
  - Low/wrong fluid
  - Worn synchronizers
  - Shift linkage misadjusted

2. Grinding When Shifting:

- **Causes:**
  - Worn synchronizers (most common)
  - Clutch dragging
  - Shift technique incorrect

3. Pops Out of Gear:

- **Causes:**
  - Worn shift forks
  - Weak detent springs
  - Worn gear dogs (engagement points)
  - Transmission mount failed

4. Noise in Neutral:

- **Causes:**
  - Worn input shaft bearing
  - Usually disappears when clutch pressed

Automatic Transmission Problems:

1. Slipping (RPM rises without acceleration):

- **Causes:**
  - Low fluid level (most common)
  - Worn clutch packs
  - Valve body problems
  - Internal damage

2. Harsh/Delayed Shifts:

- **Causes:**
  - Low fluid
  - Wrong fluid type
  - Dirty fluid

    - Solenoid failure
    - TCM issues

3. No Movement (Won't Go):

- **Causes:**
    - Very low fluid
    - Shift linkage disconnected
    - Internal failure (pump, clutches)

4. Shuddering:

- **Causes:**
    - Torque converter failure
    - Contaminated fluid
    - Worn clutches

CVT Problems:

1. Whining/Droning Noise:

- **Normal to some extent**
- **Excessive = belt/chain wear**

2. Shuddering at Low Speeds:

- **Causes:**
    - Fluid degraded
    - Belt/chain slipping
    - Contamination

3. Loss of Power:

- **Causes:**
    - Fluid level low
    - Belt/chain slipping
    - Pulley wear

Activity 2: Fluid Leak Inspection (20 min)

- Check underneath vehicles for transmission fluid leaks:
    - **Location:** Leaks appear near transmission housing
    - **Identification:**
        - Red/pink = ATF
        - Brown/amber = gear oil or old ATF

        - Green/clear = CVT fluid
    - **Common leak points:**
        - Input shaft seal (where torque converter enters)
        - Output shaft seal (where driveshaft exits)
        - Pan gasket (bottom)
        - Cooler lines (on automatics)
        - Axle seals (FWD vehicles)
- Document any leaks found
- Assess severity (drips vs puddles)

Activity 3: Shift Linkage Inspection (15 min)

- **Cable shift linkage:**
    - Check for damaged cable housing
    - Check for fraying cable
    - Check adjustment (smooth shifts?)
    - Check for rust/corrosion
- **Rod linkage:**
    - Check for bent rods
    - Check bushings for wear
    - Check for looseness
- **Symptoms of bad linkage:**
    - Hard to select gears
    - Gears don't fully engage
    - Shifter feels loose/sloppy
    - Can't select certain gears

Pass Criteria:

- ✅ Can identify symptoms of transmission problems
- ✅ Can inspect for fluid leaks
- ✅ Can check shift linkage condition
- ✅ Understands when to refer to specialist

DAY 11 ASSESSMENT:

Practical Skills Checklist:

- [ ] Identifies transmission type on vehicle
- [ ] Locates transmission components

- [ ] Checks automatic transmission fluid level correctly
- [ ] Assesses ATF condition
- [ ] Identifies fluid leaks
- [ ] Inspects shift linkage
- [ ] Recognizes symptoms requiring specialist attention

Knowledge Check (Verbal):

- What is the purpose of a transmission?
- What is a gear ratio?
- Name four types of automatic transmissions
- What color should ATF be?
- What causes transmission slipping?

DAY 12: INTEGRATION REVIEW + BASIC PM SKILLS

Learning Objectives:

- Integrate all Week 2 systems knowledge
- Understand how systems work together
- Perform basic preventive maintenance checklist
- Demonstrate cumulative skills from Weeks 1-2

MORNING SESSION: THEORY (3 Hours)

Hour 1 (9:00 AM - 10:00 AM): Powertrain Integration

How All Systems Work Together:

Complete Power Flow:

1. **Fuel System** delivers fuel
2. **Intake System** delivers air
3. **Ignition System** creates spark
4. **Engine** produces power via combustion
5. **Cooling System** removes excess heat
6. **Lubrication System** protects moving parts
7. **Emission Control** cleans exhaust gases
8. **Exhaust System** expels burned gases
9. **Clutch** connects/disconnects power

10. **Transmission** multiplies torque, adjusts speed
11. **Driveshaft/Axles** deliver power to wheels

System Interdependencies:

Engine Performance depends on:

- **Air/Fuel mixture** (intake + fuel systems)
- **Proper timing** (ignition system)
- **Optimal temperature** (cooling system)
- **Clean oil** (lubrication system)
- **Emissions system function** (proper air/fuel, catalytic converter)

Cooling System affects:

- **Engine efficiency** (must reach operating temp)
- **Emission control** (catalyst needs heat)
- **Lubrication** (oil viscosity changes with temp)
- **Cabin comfort** (heater core)

Emission Control integrates:

- **Fuel system** (proper mixture critical)
- **Ignition** (complete combustion needed)
- **Cooling** (optimal temp for catalyst)
- **Intake** (EGR recirculation)
- **Exhaust** (oxygen sensors, catalyst)
- **Engine management** (ECU coordinates all)

Example: Cold Start Sequence:

1. Turn key → Starter cranks engine
2. Fuel pump runs, builds pressure
3. ECU reads coolant temp sensor (cold)
4. ECU commands rich mixture (more fuel)
5. Ignition fires spark plugs
6. Engine starts, runs rough (cold)
7. Fast idle to warm up quickly
8. Cooling system circulates (thermostat closed initially)
9. Engine warms, idle drops to normal
10. Thermostat opens, coolant flows to radiator
11. Emissions system reaches operating temp
12. All systems now optimized

Example: Acceleration:

1. Press accelerator pedal
2. Throttle opens (more air)
3. MAF sensor measures increased airflow
4. ECU commands more fuel
5. Engine produces more power
6. Depress clutch (manual) or TC slips (automatic)
7. Select lower gear (more torque multiplication)
8. Release clutch / TC locks
9. Power transmitted through transmission
10. Vehicle accelerates
11. Cooling system removes increased heat
12. Lubrication system under higher pressure
13. Emission system processes higher exhaust flow

Hour 2 (10:00 AM - 11:00 AM): Problem Diagnosis Integration

Multi-System Problem Scenarios:

Scenario 1: Overheating Engine

- **Potential systems involved:**
  - Cooling system (primary)
  - Emission system (clogged catalyst creates back pressure)
  - Lubrication (low oil causes friction heat)
  - Ignition (retarded timing generates more heat)
- **Diagnosis approach:**
  1. Check coolant level
  2. Check for cooling system leaks
  3. Test thermostat function
  4. Check fan operation
  5. Check exhaust restriction
  6. Check oil level

Scenario 2: Black Smoke from Exhaust

- **Potential systems involved:**
  - Fuel system (over-fueling)
  - Intake system (restricted air filter)
  - Emission system (failed oxygen sensor)

    - Engine management (incorrect fuel map)
- **Diagnosis approach:**
    1. Check air filter
    2. Check oxygen sensors
    3. Check fuel pressure
    4. Scan for fault codes

Scenario 3: Poor Fuel Economy

- **Potential systems involved:**
    - Emission (oxygen sensors giving false readings)
    - Cooling (thermostat stuck open, never warms up)
    - Transmission (slipping clutches)
    - Lubrication (wrong viscosity oil, drag)
    - Intake (air leaks causing rich mixture)
- **Diagnosis approach:**
    - Check for fault codes
    - Check oxygen sensor readings
    - Check temperature gauge
    - Check air filter and intake hoses
    - Check transmission operation

Understanding Symptoms vs Root Cause:

- **Symptom:** What customer describes
- **Root cause:** Actual problem
- Example:
    - Symptom: "Car won't start"
    - Root cause could be:
        - Dead battery
        - Failed starter
        - No fuel
        - No spark
        - Timing belt broken
        - Engine seized

Systematic Approach:

1. **Gather information:** Listen to customer
2. **Verify problem:** Reproduce if possible
3. **Research:** Check service bulletins, history
4. **Isolate system:** Which system likely?

5. **Test components:** Systematic checks
6. **Identify root cause:** Not just symptoms
7. **Repair properly:** Fix cause, not just symptom
8. **Verify repair:** Test drive, recheck

Hour 3 (11:00 AM - 12:00 PM): Preventive Maintenance Philosophy

What is Preventive Maintenance?

- Scheduled service before problems occur
- Replace wear items on schedule
- Inspect for potential problems
- Adjust components as needed
- Goal: Prevent breakdowns, extend life

Why PM is Important:

- **Reliability:** Vehicle less likely to break down
- **Safety:** Worn brakes, tires caught early
- **Longevity:** Engine lasts longer with proper maintenance
- **Resale value:** Service records increase value
- **Fuel economy:** Well-maintained = efficient
- **Legal compliance:** Emission testing requirements

Manufacturer Service Intervals:

- Based on testing and experience
- Typically time or mileage (whichever first)
- Example schedule:
  - **Every 5,000 km:** Oil change, basic inspection
  - **Every 10,000 km:** Oil change, filters, full inspection
  - **Every 20,000 km:** Plus air filter, cabin filter
  - **Every 40,000 km:** Plus transmission fluid, coolant
  - **Every 60,000 km:** Major service, belts, plugs

Severe vs Normal Service:

- **Severe service:**
  - Short trips (< 10 km)
  - Extreme temperatures
  - Dusty/dirty conditions

- Towing/heavy loads
- Stop-and-go traffic
- **Requires more frequent service!**

Common PM Tasks (Overview):

Daily/Weekly Checks:

- Fluid levels (oil, coolant, brake, washer)
- Tire pressures
- Lights operation
- Wipers function
- Warning lights on dash

Oil Change Service (5,000-10,000 km):

- Engine oil and filter
- Visual inspection all fluids
- Check tire wear and pressure
- Check belts and hoses
- Check lights
- Test battery

Minor Service (10,000-20,000 km):

- Oil change service items
- Air filter inspection/replace
- Cabin filter replace
- Brake pad thickness check
- Suspension inspection
- Exhaust inspection

Major Service (40,000-60,000 km):

- Minor service items
- Transmission fluid change
- Coolant change
- Brake fluid change
- Spark plugs replace
- Timing belt replace (if due)
- Comprehensive inspection all systems

Critical Timing-Based Services:

- **Timing belt:** 60,000-100,000 km (engine damage if breaks!)
- **Coolant:** 2-5 years (corrosion protection)
- **Brake fluid:** 2-3 years (moisture absorption)
- **Transmission fluid:** 40,000-80,000 km (protection)

Documentation:

- Keep all service records
- Stamp service booklet
- File receipts
- Note mileage and date
- Helps track history
- Proves maintenance for warranty

AFTERNOON SESSION: PRACTICAL (3 Hours)

Hour 1 (1:00 PM - 2:00 PM): Comprehensive Vehicle Inspection

Activity 1: Complete Fluid Level Check (20 min)

- Students perform all fluid checks learned:
  1. **Engine oil** (Day 7, 9)
  2. **Coolant** (Day 7, 8)
  3. **Brake fluid** (Day 7)
  4. **Power steering fluid** (Day 7 - if equipped)
  5. **Transmission fluid** (Day 11 - if accessible)
  6. **Windshield washer** (always accessible)
- Document findings:
  - Level (OK, low, overfull)
  - Condition (good, marginal, bad)
  - Action needed (top up, change, monitor)

Activity 2: Under-Hood Visual Inspection (20 min)

- Systematic inspection:
  - **Belts:** Cracks, fraying, tension
  - **Hoses:** Cracks, bulges, clamps tight
  - **Battery:** Terminals clean, secure
  - **Filters:** Air filter condition
  - **Leaks:** Oil, coolant, fuel, brake fluid
  - **Wiring:** Damaged insulation, loose connectors

    - **Emission components:** Vacuum hoses, PCV valve
    - **Corrosion:** Rust, fluid deposits
- Create inspection checklist
- Mark items needing attention

Activity 3: Underside Inspection (20 min)

- Lift vehicle safely
- Inspect from underneath:
    - **Oil pan:** Leaks, damage
    - **Transmission:** Leaks at seals
    - **Exhaust system:** Rust, holes, loose hangers
    - **Suspension:** Leaking shocks, worn bushings
    - **Brake lines:** Corrosion, damage
    - **Fuel lines:** Leaks, corrosion
    - **Driveshaft/axles:** Boot tears, leaks
- Document findings with photos
- Prioritize: Safety critical vs. monitor

Pass Criteria:

- ✅ Performs all fluid checks correctly
- ✅ Conducts systematic visual inspection
- ✅ Identifies potential problems
- ✅ Documents findings professionally
- ✅ Prioritizes repairs appropriately

Hour 2 (2:00 PM - 3:00 PM): Basic PM Checklist Execution

Activity 1: Tire Inspection & Pressure Check (20 min)

- **Pressure check:**
    - Find recommended pressure (door jamb sticker)
    - Check all tires when cold
    - Adjust to specification
    - Don't forget spare tire!
- **Tread depth check:**
    - Use tread depth gauge or wear indicators
    - Minimum legal: 1.6mm
    - Replace at: 3mm (safety margin)

    - Check all four corners
- **Visual inspection:**
    - Uneven wear (alignment/suspension issue)
    - Cuts, bulges, cracks
    - Objects embedded (nails, etc.)
    - Sidewall damage

Activity 2: Light Inspection (15 min)

- Check operation of all lights:
    - **Headlights:** Low and high beam
    - **Front indicators/turn signals**
    - **Parking/position lights**
    - **Rear lights/tail lights**
    - **Brake lights** (need helper or use reflection)
    - **Reverse lights**
    - **Rear indicators/turn signals**
    - **License plate light**
    - **Hazard lights** (all four corners)
- Replace any bulbs not working
- Check for moisture in housings

Activity 3: Battery & Charging Check (15 min)

- **Visual battery check:**
    - Clean terminals (no corrosion)
    - Secure hold-down
    - No cracks in case
    - Electrolyte level (if accessible)
- **Battery voltage test:**
    - Multimeter on DC voltage
    - Red to positive, black to negative
    - **Key off:** Should read 12.4-12.8V
    - **Engine running:** Should read 13.5-14.5V
    - Below these = charging problem

Activity 4: Wiper & Washer Check (10 min)

- Test wiper operation (all speeds)
- Inspect wiper blades:
    - Cracks in rubber
    - Missing sections

    - Chattering on glass
- Test washer spray
- Top up washer fluid if needed

Pass Criteria:

- ✅ Checks and adjusts tire pressures
- ✅ Inspects tread depth
- ✅ Tests all lights
- ✅ Checks battery voltage
- ✅ Inspects wipers and tests washers

Hour 3 (3:00 PM - 4:00 PM): PM Documentation & Customer Communication

Activity 1: Creating Inspection Report (25 min)

- Students create professional inspection report
- Format should include:

Vehicle Information:

- Make, model, year
- VIN number
- Current mileage
- License plate

Fluid Levels & Condition:

- All fluids checked with status
- Any that need service

Under-Hood Inspection:

- Belt condition
- Hose condition
- Battery status
- Filter condition
- Any leaks

Under-Vehicle Inspection:

- Exhaust condition
- Suspension condition

- Any leaks
- Fluid drips

Tires & Lights:

- Tire pressures
- Tread depths
- Light operation
- Any issues

Recommendations:

- **Immediate (Red):** Safety-critical, do now
- **Soon (Amber):** Do within 1 month
- **Monitor (Green):** Check again at next service

Estimate:

- Parts needed
- Labor time
- Total cost estimate

Activity 2: Customer Communication Roleplay (20 min)

- **Scenario:** Student presents inspection findings to "customer" (instructor)
- **Practice:**
  - Explaining technical issues in simple terms
  - Prioritizing recommendations
  - Answering customer questions
  - Building trust through transparency
  - Not overselling (only recommend what's needed)

Good communication practices:

- Show customer the problems (photos, physical inspection)
- Explain consequences of delaying
- Provide options where appropriate
- Be honest about costs
- Don't use jargon
- Listen to customer concerns

Activity 3: Week 2 Cumulative Skills Test (15 min)

- Instructor calls out random tasks from Week 2:
  - "Check coolant level"

    - "Locate EGR valve"
    - "Check transmission fluid"
    - "Test clutch free play"
    - "Pressure test cooling system"
    - "Identify transmission type"
- Students must perform quickly and correctly
- Demonstrates retention of skills
- Identifies areas needing reinforcement

Pass Criteria:

- ✅ Creates comprehensive inspection report
- ✅ Communicates findings clearly to customers
- ✅ Prioritizes recommendations appropriately
- ✅ Performs Week 2 tasks on demand

DAY 12 ASSESSMENT:

Comprehensive Week 2 Evaluation:

Practical Skills (All Week 2 Content):

- [ ] Identifies all emission control components
- [ ] Performs pressure test on cooling system
- [ ] Checks all fluid levels correctly
- [ ] Assesses fluid conditions
- [ ] Identifies transmission types
- [ ] Checks transmission fluid
- [ ] Measures clutch pedal free play
- [ ] Performs comprehensive vehicle inspection
- [ ] Creates professional inspection report
- [ ] Communicates effectively with customers

Knowledge Check (Written - 30 questions):

Emissions (5 questions):

1. What does EGR reduce?
2. Name three functions of catalytic converter
3. What stores fuel vapors?
4. What is BS6 standard?

5. Why must PCV system work?

Cooling (5 questions): 6. Why pressurize cooling system? 7. What does thermostat do? 8. Optimal engine temperature range? 9. Name three causes of overheating 10. When does cooling fan run?

Lubrication (5 questions): 11. Name five functions of engine oil 12. What does 5W-30 mean? 13. Why is API rating important? 14. How often change oil? 15. What causes oil to turn milky?

Clutch (5 questions): 16. What is clutch's main function? 17. Name three main clutch components 18. What is friction point? 19. Symptom of slipping clutch? 20. What does excessive free play cause?

Transmission (5 questions): 21. Why do we need transmission? 22. What is gear ratio? 23. Difference between CVT and automatic? 24. What color is ATF? 25. How often check transmission fluid?

Integration (5 questions): 26. How do cooling and emission systems work together? 27. What systems involved in cold start? 28. Why is PM important? 29. What is "severe service"? 30. Name three daily checks.

Pass Mark: 70% (21/30)

WEEK 2 SUMMARY & OUTCOMES

Week 2 Learning Journey: "Understanding Engine Support Systems & Power Delivery"

Day 7: Emission control, basic fluid checking **Day 8:** Cooling system, pressure testing, coolant service **Day 9:** Lubrication system, oil properties, oil service basics **Day 10:** Clutch system, hydraulics, adjustment understanding **Day 11:** Transmission types, fluid checking, problem recognition **Day 12:** System integration, preventive maintenance, comprehensive inspection

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Week 2 Skills Achieved:

✅ **Emission System Skills:**

- Identifies emission components
- Understands environmental responsibility
- Recognizes emission system problems

✅ Cooling System Skills:

- Traces coolant flow path
- Performs pressure testing
- Checks coolant level and condition
- Recognizes overheating causes

✅ Lubrication Skills:

- Checks oil level and condition
- Understands oil specifications
- Selects correct oil for vehicle
- Knows oil change procedure (observation)

✅ Clutch Skills:

- Identifies clutch components
- Measures pedal free play
- Checks clutch fluid
- Locates friction point
- Recognizes clutch problems

✅ Transmission Skills:

- Identifies transmission types
- Checks automatic transmission fluid
- Understands gear ratios
- Recognizes transmission problems
- Inspects for leaks

✅ Integration & PM Skills:

- Performs comprehensive vehicle inspection
- Checks all fluid levels
- Creates professional inspection reports
- Communicates findings to customers
- Understands preventive maintenance importance